

SERANGOON JUNIOR COLLEGE

H2 MATHEMATICS PAPER 1

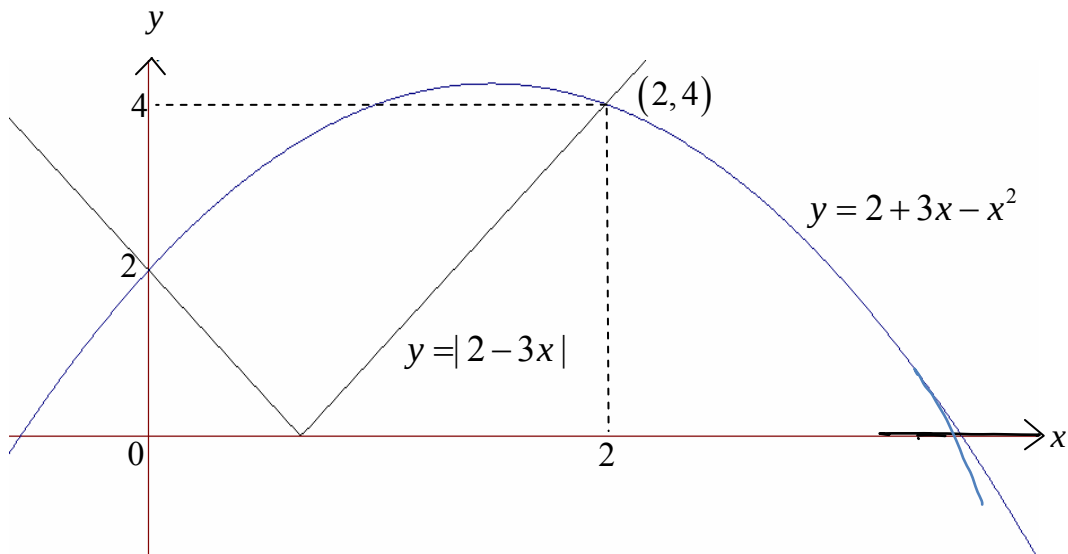
MARKING SCHEME

Qn No

1. Solve $|2 - 3x| > 2 + 3x - x^2$.

Solution:

Method 1



Intersection points – **G2** (1 mark for each intersection)

Shape and labeling of graphs and axes – **G1**

From the sketch,

$x < 0$ or $x > 2$

Method 2 (Algebraic Method)

For $x \geq \frac{2}{3}$,

$$-(2 - 3x) > 2 + 3x - x^2$$

$$x^2 - 4 > 0$$

$$(x - 2)(x + 2) > 0$$

$$\therefore x > 2 \text{ or } x < -2 \left(\text{rejected } \because x \geq \frac{2}{3} \right)$$

$$\text{For } x < \frac{2}{3},$$

$$2 - 3x > 2 + 3x - x^2$$

$$x^2 - 6x > 0$$

$$x(x - 6) > 0$$

$$x < 0 \text{ or } x > 6 \left(\text{rejected } \because x < \frac{2}{3} \right)$$

$$\text{So } x < 0 \text{ or } x > 2.$$

- 2(a) With respect to an origin O , the points A , B and C have position vectors $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$, $14\mathbf{i} + \mathbf{k}$ and $2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ respectively.

The point P on AB is such that $AP : PB = 1 : \lambda$, $\lambda > 0$.

Given that $\overrightarrow{CP}$ is parallel to the vector $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$,

find λ .

- (b) The angle between the vector $\mathbf{i} + \mu\mathbf{k}$ and the vector $\mathbf{j} + 2\mathbf{k}$ is 150° . Find the exact value of the constant μ .

Solution:

- (a)

$$\overrightarrow{OP} = \frac{\overrightarrow{OB} + \lambda \overrightarrow{OA}}{1 + \lambda} = \frac{\begin{pmatrix} -14 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}}{1 + \lambda} = \frac{1}{1 + \lambda} \begin{pmatrix} -14 + 2\lambda \\ 4\lambda \\ 1 + \lambda \end{pmatrix}$$

$$\overrightarrow{CP} = \frac{1}{1 + \lambda} \begin{pmatrix} -14 + 2\lambda \\ 4\lambda \\ 1 + \lambda \end{pmatrix} - \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{-14 + 2\lambda}{1 + \lambda} - 2 \\ \frac{4\lambda}{1 + \lambda} + 5 \\ 2 \end{pmatrix}$$

$$\overrightarrow{CP} = \begin{pmatrix} \frac{-14 + 2\lambda}{1 + \lambda} - 2 \\ \frac{4\lambda}{1 + \lambda} + 5 \\ 2 \end{pmatrix} = k \begin{pmatrix} -2 \\ 4 \\ 1 \end{pmatrix}$$

$$\Rightarrow k = 2$$

$$\frac{4\lambda}{1 + \lambda} + 5 = 4k = 8 \quad \text{or} \quad \frac{-14 + 2\lambda}{1 + \lambda} - 2 = -2k = -4$$

$$\Rightarrow \lambda = 3$$

(b)

$$\cos 150^\circ = \frac{\begin{pmatrix} 1 \\ 0 \\ \mu \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}}{\sqrt{5} \sqrt{\mu^2 + 1}}$$

$$1 - \frac{\sqrt{3}}{2} = \frac{2\mu}{\sqrt{5} \sqrt{\mu^2 + 1}}$$

$$14\mu = -\sqrt{15} \sqrt{\mu^2 + 1}$$

$$116\mu^2 = 15(\mu^2 + 1)$$

$$11\mu^2 = 15$$

$$11\mu = -\sqrt{15} \text{ (reject positive value of } \mu \text{)}$$

3

Show that the equation $y = \frac{4x^2 + 14x + 11}{2(x+1)}$ can be written as $y = 2x + A + \frac{B}{2(x+1)}$ where

A and B are constants to be determined.

Hence, state the sequence of transformations which transform the graph of $y = x + \frac{1}{x}$ to

the graph of $y = \frac{4x^2 + 14x + 11}{2(x+1)}$.

Solution:

$$y = \frac{4x^2 + 14x + 11}{2(x+1)}$$

$$y = \frac{2x(2x+2) + 10x + 11}{2(x+1)}$$

$$y = 2x + \frac{10x + 11}{2(x+1)}$$

$$y = 2x + \frac{5(2x+2) + 1}{2(x+1)}$$

$$y = 2x + 5 + \frac{1}{2(x+1)}$$

$$A = 5, B = 1$$

Method 2: By long division

$$\begin{array}{r}
 2x+5 \\
 2(x+1) \overline{) 4x^2 + 14x + 11} \\
 \underline{-(4x^2 + 4x)} \\
 10x + 11 \\
 \underline{-(10x + 10)} \\
 1
 \end{array}$$

$$y = 2x + 5 + \frac{1}{2(x+1)}$$

$$\text{From } y = 2x + 5 + \frac{1}{2(x+1)}$$

$$y = 2(x+1) + \frac{1}{2(x+1)} + 3$$

$$\begin{aligned}
 f(x) = x + \frac{1}{x} &\xrightarrow{f(2x)} 2x + \frac{1}{2x} \xrightarrow{f(x+1)} 2(x+1) + \frac{1}{2(x+1)} \\
 &\xrightarrow{f'(x)+3} 2(x+1) + \frac{1}{2(x+1)} + 3
 \end{aligned}$$

Scaling parallel to x-axis by a factor of $\frac{1}{2}$

Translation of 1 unit in the negative x-direction

Translation of 3 units in the positive y-direction

Alternative solution (1):

Translation of 3 units in the positive y-direction

Scaling parallel to x-axis by a factor of $\frac{1}{2}$

Translation of 1 unit in the negative x-direction

Alternative solution (2):

Translation of 2 unit in the negative x-direction

Scaling parallel to x-axis by a factor of $\frac{1}{2}$

Translation of 3 units in the positive y-direction

Alternative solution (3):

Translation of 3 units in the positive y-direction

Translation of 2 unit in the negative x-direction

Scaling parallel to x-axis by a factor of $\frac{1}{2}$

4 Prove by induction that

$$(2^2)(1) + (3^2)(2) + (4^2)(2^2) + (5^2)(2^3) + \dots + (n+1)^2(2^{n-1}) = (n^2 + 2)2^n - 2$$

for all positive integers n .

Solution

Let P_n denote the statement

$$\sum_{r=1}^n (r+1)^2(2^{r-1}) = (n^2 + 2)2^n - 2$$

$$\text{When } n = 1 : \quad LHS = (2^2)(1) = 4$$

$$RHS = (1^2 + 2)(2) - 2 = 4 = LHS$$

Hence P_1 is true

Assume that P_k is true for some $n = k$. i.e.

$$\begin{aligned}\sum_{r=1}^k (r+1)^2 (2^{r-1}) &= (k^2 + 2)2^k - 2 \\ \sum_{r=1}^{k+1} (r+1)^2 (2^{r-1}) &= (k^2 + 2)2^k - 2 + (k+2)^2 (2^k) \\ &= 2^k (k^2 + 2 + k^2 + 4k + 4) - 2 \\ &= (k^2 + 2k + 3)2^{k+1} - 2 \\ &= [(k+1)^2 + 2]2^{k+1} - 2\end{aligned}$$

P_k is true $\Rightarrow P_{k+1}$ is true.

P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true. Hence by mathematical induction, P_n is true for all positive integers n .

- 5 Three students, Sally (S), Preeti (P) and Chunli (C) have been tasked to build 80 models for sale during their college's fundraising campaign. Three steps J, K and L are required in the building of each model. Due to the different proficiency levels of the students, the time that they take to complete each step is different. The following table provides the breakdown of time that each student takes to complete each step in seconds:

	J (sec)	K (sec)	L(sec)
S	210	30	10
P	100	15	7
C	350	40	12

The total time taken by the students to complete the 80 models is 21240 seconds while the total time taken for the students to complete step **L** is 790 seconds. Find the number of models that each student builds.

Solution

Let x , y and z be the number of models that S, P and C build respectively.

$$x + y + z = 80$$

$$10x + 7y + 12z = 790$$

$$250x + 122y + 402z = 21240$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 80 \\ 10 & 7 & 12 & 790 \\ 250 & 122 & 402 & 21240 \end{array} \right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 35 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 25 \end{array} \right)$$

$$x = 35, y = 20, z = 25$$

Sally, Preeti and Chunli build 35, 20 and 25 models respectively.

- 6 The functions f and g are defined as follows:

$$f: x \rightarrow 3 - 2e^{x+1}, \quad x \in \mathbb{R}$$

$$g: x \rightarrow x^2 + 2x + 2, \quad x \in \mathbb{R}, x \leq -1.$$

- a Find g^{-1} , giving its domain.
- b Explain why the composite function fg exists. Find fg in similar form and state its exact range.

Solution:

- a Let $y = (x+1)^2 + 1$
 $\Rightarrow x = -1 \pm \sqrt{y-1}$

Since $x \leq -1$,

$$\therefore x = -1 - \sqrt{y-1}$$

$$\therefore g^{-1}(x) = -1 - \sqrt{x-1}, \quad x \geq 1$$

b

$$R_g = [1, \infty)$$

$$D_f = \mathbb{R}$$

$$R_g \subseteq D_f,$$

$\therefore fg$ exists

$$fg(x) = 3 - 2e^{(x+1)^2+2}; \quad x \in \mathbb{R}, \quad x \leq -1.$$

$$R_{fg} = (-\infty, 3 - 2e^2]$$

7

The curve C has equation

$$y = \frac{x^2 + b}{x + a}, \text{ where } a > 0 \text{ and } b > 0.$$

- (i) State the coordinates of the intersection(s) of the curve C with the axes in terms of a and b .
- (ii) Find the equation(s) of the asymptote(s).
- (iii) Draw a sketch of the curve C , labeling the equation(s) of its asymptote(s) and coordinates of any intersection with the axes.
- (iv) Hence find the range of values of k , where k is a positive constant, for which the equation $x^2 + b = (x + a)(kx - a)$ has no real root.

Solution:

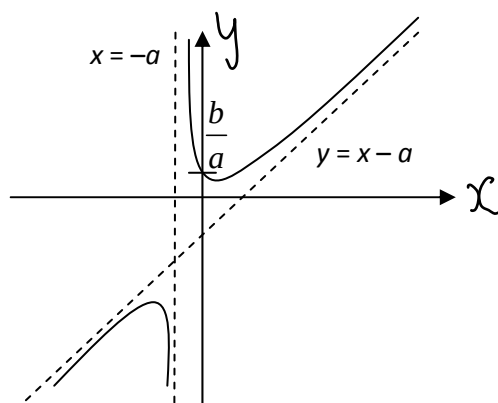
- (i) $(0, \frac{b}{a})$

(ii)
$$y = \frac{x^2 + b}{x + a}$$

$$= (x - a) + \frac{b + a^2}{x + a}$$

$x = -a$; $y = x - a$ are equations of the asymptotes.

(iii)



(iv)
$$kx - a = \frac{x^2 + b}{x + a}$$

$$\Rightarrow x^2 + b = (x + a)(kx - a)$$

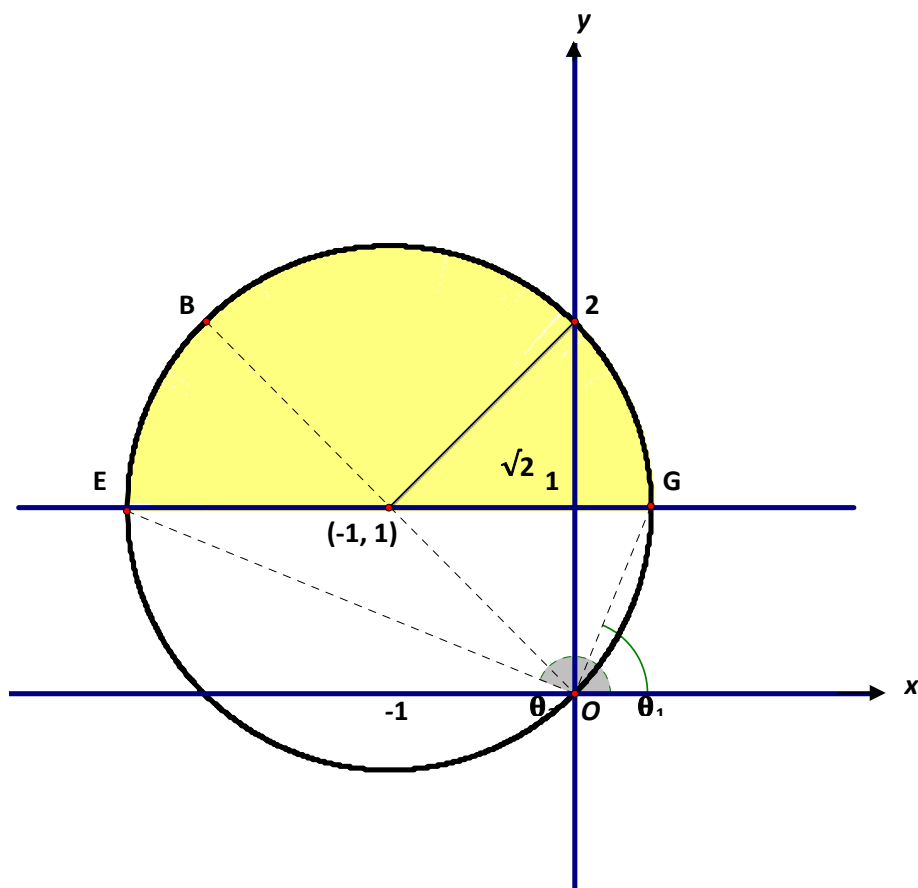
From graph, $0 < k \leq 1$

8 Sketch on an Argand diagram, the set of points representing all complex numbers z such that

$$|z + 1 - i| \leq |1 + i| \text{ and } |z| \geq |z - 2i|.$$

Hence find the range of $|z|$ and $\arg(z)$.

Solution:



Let C be the centre of the circle.

$$\text{Min } |z| = 1$$

$$\text{Max } |z| = OB = OC + CB = 2\sqrt{2}$$

$$1 \leq |z| \leq 2\sqrt{2}$$

$$\text{Pt E } (-1 - \sqrt{2}, 1), \arg(-1 - \sqrt{2} + i) = 2.75$$

$$\text{Pt G } (-1 + \sqrt{2}, 1), \arg(-1 + \sqrt{2} + i) = 1.18$$

$$1.18 \leq \arg(z) \leq 2.75$$

9a

Use the substitution $x = 2 \tan \theta$ to find the exact value of $\int_{\frac{2}{\sqrt{3}}}^2 \frac{1}{x\sqrt{x^2+4}} dx$.

Solution:

$$x = 2 \tan \theta, \quad \frac{dx}{d\theta} = 2 \sec^2 \theta$$

$$\text{When } x = 2, \quad \theta = \frac{\pi}{4}$$

$$\text{When } x = \frac{2}{\sqrt{3}}, \quad \theta = \frac{\pi}{6}$$

$$\int_{\frac{2}{\sqrt{3}}}^2 \frac{1}{x\sqrt{x^2+4}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{(2 \tan \theta)\sqrt{4 \tan^2 \theta + 4}} (2 \sec^2 \theta) d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{(2 \tan \theta)\sqrt{4(\tan^2 \theta + 1)}} (2 \sec^2 \theta) d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{(2 \tan \theta)2\sqrt{\sec^2 \theta}} (2 \sec^2 \theta) d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{2} \frac{1}{\tan \theta} (\sec \theta) d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\cos \theta}{\sin \theta} \left(\frac{1}{\cos \theta} \right) d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{\sin \theta} d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \operatorname{cosec} \theta d\theta$$

$$\begin{aligned}
&= -\frac{1}{2} \left[\ln \left| \operatorname{cosec} \theta + \cot \theta \right| \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} \\
&= -\frac{1}{2} \left[\ln \left| \operatorname{cosec} \frac{\pi}{4} + \cot \frac{\pi}{4} \right| - \ln \left| \operatorname{cosec} \frac{\pi}{6} + \cot \frac{\pi}{6} \right| \right] \\
&= -\frac{1}{2} \left[\ln \frac{\sqrt{2} + 1}{2 + \sqrt{3}} \right]
\end{aligned}$$

- b (i) By writing $x = A(1-x) + B$, where A and B are constants, prove that

$$\int \frac{x}{\sqrt{1-x}} dx = \frac{2}{3}(1-x)^{\frac{3}{2}} - 2(1-x)^{\frac{1}{2}} + C$$

where C is an arbitrary constant.

(ii) Hence, find $\int \sqrt{x} (\sin^{-1} \sqrt{x}) dx$

- b (i) $x = A(1-x) + B$
By inspection, $A = -1$, $B = 1$
(Or Compare coefficient)

$$\begin{aligned}
\int \frac{x}{\sqrt{1-x}} dx &= \int \frac{-(1-x) + 1}{\sqrt{1-x}} dx \\
&= \int -\sqrt{1-x} dx + \int \frac{1}{\sqrt{1-x}} dx + C
\end{aligned}$$

$$= -\frac{(1-x)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)(-1)} + \frac{(1-x)^{\frac{1}{2}}}{(-1)\left(\frac{1}{2}\right)} + C$$

$$= \frac{2}{3}(1-x)^{\frac{3}{2}} - 2(1-x)^{\frac{1}{2}} + C$$

$$(ii) \int \sqrt{x} (\sin^{-1} \sqrt{x}) dx$$

$$\frac{dV}{dx} = \sqrt{x} \Rightarrow V = \frac{2}{3} x^{\frac{3}{2}}$$

$$u = \sin^{-1} \sqrt{x} \Rightarrow \frac{du}{dx} = \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}}$$

$$\int \sqrt{x} \sin^{-1} \sqrt{x} dx$$

$$= \frac{2}{3} x^{\frac{3}{2}} \sin^{-1} \sqrt{x} - \int \frac{2}{3} x^{\frac{3}{2}} \cdot \frac{1}{2\sqrt{x}} \cdot \frac{1}{\sqrt{1-x}} dx$$

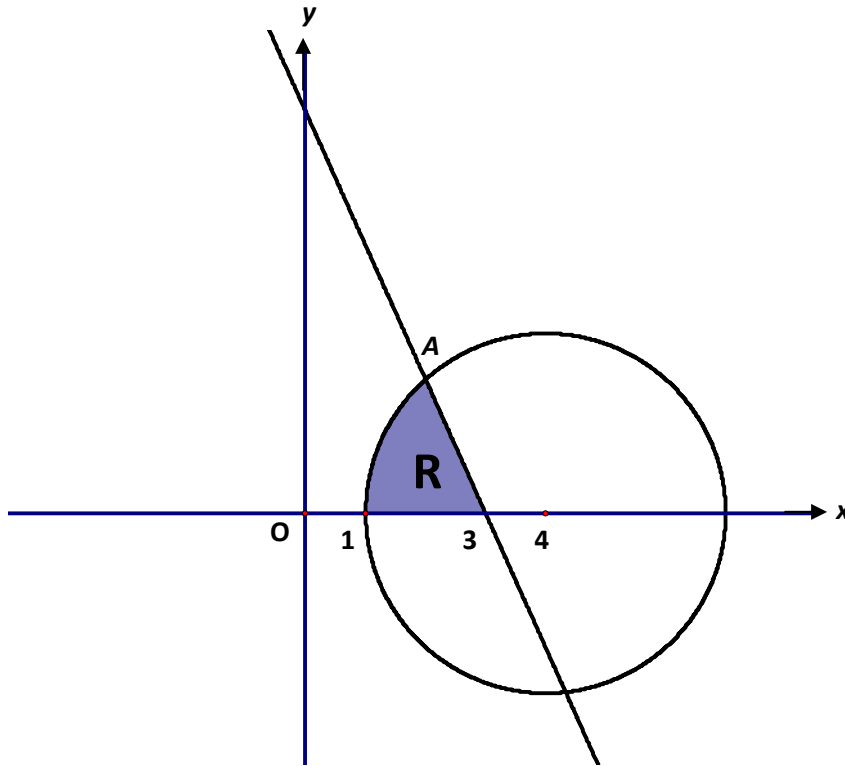
$$= \frac{2}{3} x^{\frac{3}{2}} \sin^{-1} \sqrt{x} - \frac{1}{3} \int \frac{x}{\sqrt{1-x}} dx$$

$$= \frac{2}{3} x^{\frac{3}{2}} \sin^{-1} \sqrt{x} - \frac{1}{3} \left[\frac{2}{3} (1-x)^{\frac{3}{2}} - 2(1-x)^{\frac{1}{2}} + C \right]$$

$$= \frac{2}{3} x^{\frac{3}{2}} \sin^{-1} \sqrt{x} - \frac{2}{9} (1-x)^{\frac{3}{2}} + \frac{2}{3} (1-x)^{\frac{1}{2}} + D$$

10

The diagram shows the region **R** bounded by the circle with equation $(x-4)^2 + y^2 = 9$, the straight line $y = -\sqrt{5}x + 3\sqrt{5}$ and the x -axis. The coordinates of A, which is the intersection point of the line $y = -\sqrt{5}x + 3\sqrt{5}$ and the circle, is $(2, \sqrt{5})$.



- (a) Find the area of the region **R**, giving your answer correct to 3 decimal places.
- (b) Show that the volume, S , formed when **R** is rotated through 2π radians about the y -axis is

$$S = \pi \int_0^{\sqrt{5}} \left[\frac{(y - 3\sqrt{5})^2}{5} - (25 - y^2 - 8\sqrt{9 - y^2}) \right] dy.$$

[There is no need to evaluate S .]

Solution:

- (a) Area of region **R**
- $$= \int_1^2 \sqrt{9 - (x-4)^2} dx + \int_2^3 (-\sqrt{5}x + 3\sqrt{5}) dx$$
- $$= 1.5487 + 1.1180$$

$$= 2.667 \text{ (3d.p)}$$

$$(b) \quad (x-4)^2 + y^2 = 9$$

$$(x-4)^2 = 9 - y^2$$

$$x-4 = \pm\sqrt{9-y^2}$$

$$x = 4 \pm \sqrt{9-y^2}$$

$$\text{Since } x < 4, \quad x = 4 - \sqrt{9-y^2}$$

$$y = -\sqrt{5}x + 3\sqrt{5}$$

$$-\sqrt{5}x = y - 3\sqrt{5}$$

$$x = \frac{y-3\sqrt{5}}{-\sqrt{5}}$$

$$\text{Volume, } S = \pi \int_0^{\sqrt{5}} \left(\frac{y-3\sqrt{5}}{-\sqrt{5}} \right)^2 - \left(4 - \sqrt{9-y^2} \right)^2 dy$$

$$= \pi \int_0^{\sqrt{5}} \frac{(y-3\sqrt{5})^2}{5} - \left[16 - 8\sqrt{9-y^2} + (9-y^2) \right] dy$$

$$= \pi \int_0^{\sqrt{5}} \frac{(y-3\sqrt{5})^2}{5} - (25 - y^2 - 8\sqrt{9-y^2}) dy$$

- 11(i) Given that $y = \ln\left(\frac{\cos x}{e-x}\right)$, show that $\frac{dy}{dx} = -\tan x + \frac{1}{(e-x)}$. By further differentiation of this result, or otherwise, find the Maclaurin's series of y up to and including the term in x^3 .

- (ii) Deduce the Maclaurin's series for the equation $y = \ln\left(\frac{\cos(ex)}{1+x}\right)$ up to and including the term in x^3 .

Solution:

$$y = \ln\left(\frac{\cos x}{e-x}\right) = \ln(\cos x) - \ln(e-x)$$

$$\frac{dy}{dx} = \frac{-\sin x}{\cos x} + \frac{1}{e-x} = -\tan x + \frac{1}{e-x} \text{ (Shown)}$$

Method 1

$$\frac{d^2y}{dx^2} = -\sec^2 x + \frac{1}{(e-x)^2}$$

$$\frac{d^3y}{dx^3} = -2\sec^2 x \tan x + \frac{2}{(e-x)^3}$$

Method 2

$$(e-x) \frac{dy}{dx} = -(e-x) \tan x + 1$$

$$(e-x) \frac{d^2y}{dx^2} - \frac{dy}{dx} = (x-e) \sec^2 x + \tan x$$

$$(e-x) \frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} - \frac{d^2y}{dx^2} = 2(x-e) \sec^2 x \tan x + \sec^2 x + \sec^2 x$$

$$(e-x) \frac{d^3y}{dx^3} - 2 \frac{d^2y}{dx^2} = 2 \sec^2 x [(x-e) \tan x + 1]$$

When $x = 0$,

$$y = \ln\left(\frac{1}{e}\right) = -\ln e = -1$$

$$\frac{dy}{dx} = \frac{1}{e}$$

$$\frac{d^2y}{dx^2} = \frac{1}{e^2} - 1$$

$$\frac{d^3y}{dx^3} = \frac{2}{e^3}$$

So the Maclaurin Expansion of y is

$$y = -1 + \frac{1}{e}x + \frac{\left(\frac{1}{e^2} - 1\right)}{2!}x^2 + \frac{\left(\frac{2}{e^3}\right)}{3!}x^3 + \dots$$

$$= -1 + \frac{1}{e}x + \frac{1}{2}\left(\frac{1}{e^2} - 1\right)x^2 + \frac{1}{3e^3}x^3 + \dots$$

(ii) Replace x by $-ex$

$$\text{Then } y = \ln\left(\frac{\cos(-ex)}{e - (-ex)}\right) = \ln\left(\frac{\cos(ex)}{e(1+x)}\right) = \ln\left(\frac{\cos(ex)}{(1+x)}\right) + \ln\frac{1}{e}$$

$$\ln\left(\frac{\cos(ex)}{(1+x)}\right) + \ln\frac{1}{e} = -1 + \frac{1}{e}(-ex) + \frac{1}{2}\left(\frac{1}{e^2} - 1\right)(-ex)^2 + \frac{1}{3e^3}(-ex)^3 + \dots$$

$$\Rightarrow \ln\left(\frac{\cos(ex)}{(1+x)}\right) = -x + \frac{1}{2}\left(\frac{1}{e^2} - 1\right)(ex)^2 - \frac{1}{3e^3}(ex)^3 + \dots$$

$$\ln\left(\frac{\cos(ex)}{(1+x)}\right) = -x + \frac{1}{2}(1 - e^2)x^2 - \frac{1}{3}x^3 + \dots$$

12(a) The variables x and y are related by the differential equation $x \frac{dy}{dx} - y + \frac{2}{x} = 0$.

Show that by means of the substitution $y = ux$, the differential equation may be reduced to $\frac{du}{dx} = -\frac{2}{x^3}$.

Hence, or otherwise, find the general solution for y in terms of x .

Sketch, for $x > 0$, two members of the family of solution curves corresponding to a positive value and a negative value of the arbitrary constant.

Solution:

12(a) Given $x \frac{dy}{dx} - y + \frac{2}{x} = 0 \dots (1)$

Let $y = ux$

$$\frac{dy}{dx} = u + x \frac{du}{dx}$$

Sub into eq (1)

$$x \left(u + x \frac{du}{dx} \right) - ux + \frac{2}{x} = 0$$

$$xu + x^2 \frac{du}{dx} - ux + \frac{2}{x} = 0$$

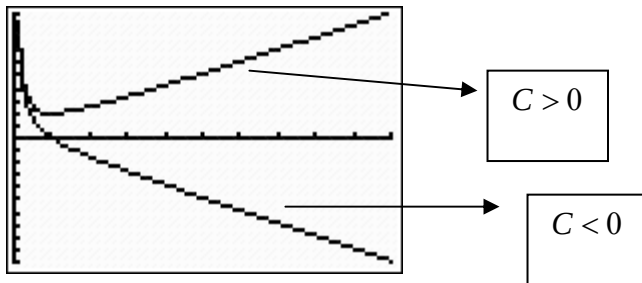
$$\frac{du}{dx} = -\frac{2}{x^3} \text{ (Shown)}$$

$$u = \int -\frac{2}{x^3} dx$$

$$u = -2 \left[\frac{x^{-3+1}}{-2} \right] + C$$

$$\frac{y}{x} = \frac{1}{x^2} + C$$

$$y = \frac{1}{x} + Cx$$



- (b) The rate at which a chemical evaporates at room temperature is proportional to the amount of substance which has not yet evaporated. If the initial amount of chemical was 500g and the amount which has evaporated at time t is x . Write down a differential equation involving x and t . Show that $x = 500 - Ae^{-kt}$, where A and k are constants.

Show that the time taken for half the chemical to evaporate is $\frac{1}{k} \ln 2$.

Sol

$$\frac{dx}{dt} = a(500 - x)$$

$$\int \frac{1}{500-x} dx = \int a dt$$

$$-\ln|500-x| = at + C$$

$$\ln|500-x| = -at - C$$

$$500-x = e^{-at-C}$$

$$x = 500 - Ae^{-at}, A = e^{-C}$$

$$x = 500 - Ae^{-kt}, k = a \text{ (Shown)}$$

$$\text{When } t = 0, x = 0$$

$$0 = 500 - A$$

$$A = 500$$

$$\therefore x = 500 - 500e^{-kt}$$

$$\text{When } x = 250 \text{ (Half the chemical)}$$

$$250 = 500 - 500e^{-kt}$$

$$500e^{-kt} = 250$$

$$e^{-kt} = \frac{250}{500} = \frac{1}{2}$$

$$-kt = \ln \frac{1}{2}$$

$$t = -\frac{1}{k} \ln \frac{1}{2}$$

$$t = \frac{1}{k} \ln 2 \text{ (shown)}$$

- 13 It is given that p, q and v are the first three terms of a geometric progression. It is also given that p, v and q are the first, third and fourth term of an arithmetic progression respectively.

- (i) Show that

$$q^2 = pv \quad \text{and} \quad v = \frac{2q+p}{3}$$

- (ii) Hence show that $3\left(\frac{q}{p}\right)^2 - 2\left(\frac{q}{p}\right) - 1 = 0$.

- (iii) Find the sum to infinity in terms of p for the geometric progression.

Solution:

- (i) Let the common difference of the AP be d

$$q - v = d$$

$$v - p = 2d$$

$$\therefore 2(q - v) = (v - p)$$

$$v = \frac{2q + p}{3} \dots\dots\dots(1)$$

Let the common ratio of the GP be R

$$\frac{v}{q} = R$$

$$\frac{q}{p} = R$$

$$\therefore \frac{v}{q} = \frac{q}{p}$$

$$q^2 = pv \dots\dots\dots(2)$$

- (ii) Sub (1) into (2)

$$q^2 = p\left(\frac{2q + p}{3}\right)$$

Divide by q^2 throughout

$$\frac{q^2}{p^2} = \left(\frac{1}{3} + \frac{2q}{3p}\right)$$

$$3\left(\frac{q}{p}\right)^2 - 2\left(\frac{q}{p}\right) - 1 = 0$$

- (iii) Ratio of the geometric progression, $R = \frac{q}{p}$

From equation in ii),

$$\frac{q}{p} = -\frac{1}{3} \text{ or } 1(\text{rej})$$

$$\begin{aligned} \text{Sum to infinity} &= \frac{p}{1 - \left(-\frac{1}{3}\right)} \\ &= \frac{3p}{4} \end{aligned}$$

- 14a. Given $f(r) = \frac{1}{r^2}$, $r \neq 1$, show that

$$f(r-1) - f(r) = \frac{2r-1}{r^2(r-1)^2}.$$

- (i) Hence or otherwise, find the sum to infinity of the

$$\text{series } \frac{3}{2^2(1)^2} + \frac{5}{3^2(2)^2} + \frac{7}{4^2(3)^2} + \dots$$

- (ii) Deduce that $\sum_{r=2}^n \frac{2r-1}{r^2(r+1)^2} < 1$.

- 14b. The sequence of positive numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence relation

$$x_{n+1} = 4 - \frac{5}{x_n + 2} \text{ for } n \geq 1.$$

- (i) As $n \rightarrow \infty$, $x_n \rightarrow \alpha$. Find the exact value of α .

- (ii) Show that if $x_n > \alpha$, then $x_{n+1} < x_n$.
- (iii) Use a calculator to determine the behavior of the sequence $\{x_n\}$ when $x_1 = 4$. Hence state briefly how the results in (ii) relate to the behavior of the sequence.

Solution:

14a.

$$f(r-1) - f(r) = \frac{1}{(r-1)^2} - \frac{1}{r^2}, \quad r \neq 1$$

$$\frac{1}{(r-1)^2} - \frac{1}{r^2}$$

$$= \frac{r^2 - r^2 + 2r - 1}{r^2 (r-1)^2}$$

$$= \frac{2r-1}{r^2 (r-1)^2}$$

$$(i) \quad \frac{3}{2^2(1)^2} + \frac{5}{3^2(2)^2} + \frac{7}{4^2(3)^2} + \dots = \sum_{r=2}^{\infty} \frac{2r-1}{r^2 (r-1)^2}$$

Consider:

$$\begin{aligned} \sum_{r=2}^n f(r-1) - f(r) &= \frac{1}{1^2} - \cancel{\frac{1}{2^2}} \\ &\quad + \cancel{\frac{1}{2^2}} - \frac{1}{3^2} \\ &\quad + \cancel{\frac{1}{3^2}} - \frac{1}{4^2} \\ &\quad + \dots \\ &\quad + \cancel{\frac{1}{(n-2)^2}} - \cancel{\frac{1}{(n-1)^2}} \end{aligned}$$

$$+\frac{1}{(n-1)^2}-\frac{1}{n^2}$$

$$=1-\frac{1}{n^2}$$

$$\sum_{r=2}^{\infty} \frac{2r-1}{r^2(r-1)^2} = 1 \text{ as } n \rightarrow \infty$$

$$(ii) \text{ Since } \frac{2r-1}{r^2(r+1)^2} < \frac{2r-1}{r^2(r-1)^2},$$

$$\sum_{r=2}^n \frac{2r-1}{r^2(r+1)^2} < \sum_{r=2}^n \frac{2r-1}{r^2(r-1)^2} = 1 - \frac{1}{n^2} < 1 \text{ (since } 0 < \frac{1}{n^2} < 1)$$

$$\sum_{r=2}^n \frac{2r-1}{r^2(r+1)^2} < 1$$

$$(b) \quad (i) \quad x_{n+1} = 4 - \frac{5}{x_1 + 2}.$$

When $n \rightarrow \infty$, x_{n+1} and $x_n \rightarrow \alpha$

$$\alpha = 4 - \frac{5}{\alpha + 2}$$

$$\alpha(\alpha + 2) = 4(\alpha + 2) - 5$$

$$\alpha^2 + 2\alpha = 4\alpha + 8 - 5$$

$$\alpha^2 - 2\alpha - 3 = 0$$

$$(\alpha + 1)(\alpha - 3) = 0$$

$$\alpha = -1 \text{ or } \alpha = 3$$

(rej as x_n is positive)

$$\begin{aligned}
\text{(ii)} \quad x_{n+1} - x_n &= 4 - \frac{5}{x_n + 2} - x_n \\
x_{n+1} - x_n &= - \left(\frac{x_n^2 - 2x_n - 3}{x_n + 2} \right) \\
x_{n+1} - x_n &= - \left(\frac{(x_n + 1)(x_n - 3)}{x_n + 2} \right)
\end{aligned}$$

$$\text{Since } x_n > 3, \left(\frac{(x_n + 1)(x_n - 3)}{x_n + 2} \right) > 0$$

$$x_{n+1} - x_n = \left(\frac{x_n + 1)(x_n - 3)}{x_n + 2} \right) < 0$$

$$x_{n+1} < x_n$$

Alternative solution (1):

$$\text{(ii)} \quad x_n - x_{n+1} = x_n - \left(4 - \frac{5}{x_n + 2} \right)$$

$$x_n - x_{n+1} = x_n - 4 + \frac{5}{x_n + 2}$$

$$x_n - x_{n+1} = \frac{(x_n + 1)(x_n - 3)}{x_n + 2}$$

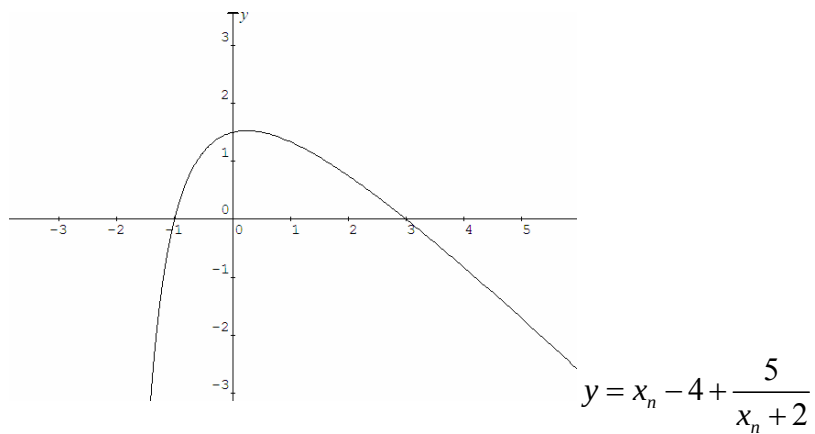
$$\text{Since } x_n > 3, \left(\frac{(x_n + 1)(x_n - 3)}{x_n + 2} \right) > 0$$

$$x_n - x_{n+1} > 0$$

$$x_{n+1} < x_n$$

Alternative solution (2):

use of GC to sketch $y = x_n - 4 + \frac{5}{x_n + 2}$

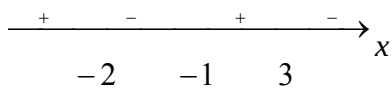


From the graph, $x_n > 3$, $y = x_n - 4 + \frac{5}{x_n + 2} < 0$

$$x_{n+1} - x_n < 0$$

$$x_{n+1} < x_n$$

Consider $x_{n+1} - x_n = -\left(\frac{(x_n + 1)(x_n - 3)}{x_n + 2}\right)$



(iii) It decreases and converges to 3.

As $x_1 = 4 > (\alpha = 3)$, from (ii) the sequence will decrease and eventually converges to 3.

